

Yahir Aguirre Garcia

Mechatronics Engineer | Mechanical Design · Automation · Robotics

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Profile

Mechatronics engineer with hands-on experience in the design, analysis, and simulation of electrical and control systems. Have taken 6+ projects from concept to functional physical prototype, integrating mechanical design, automation, and robotics. Passionate about applied engineering, continuous learning, and building efficient solutions for technological and industrial projects.

Education

Universidad Popular Autónoma del Estado de Puebla (UPAEP), Bachelor's in Mechatronics Engineering – Puebla, Mexico Aug 2020 – present

- GPA: 9.16 / 10.0.

Universidad del Valle de Puebla, Technical High-School Degree in Mechatronics – Puebla, Mexico Oct 2017 – May 2020

Languages

English: Advanced — TOEFL ITP 590.

Spanish: Native.

Technical Skills

Mechanical Design (CAD): SolidWorks · CATIA · Siemens NX · AutoCAD — solid modeling, assemblies, motion studies, dimensioning, and GD&T.

Manufacturing (CAM): Esprit — CNC machining programming.

Manufacturing & Workshop: Manual lathe · milling · welding · drilling · metal and woodworking · benchtop tooling.

Simulation & Analysis: COMSOL Multiphysics · MATLAB / Simulink.

Automation & Control: CODESYS · CADe SIMU · Factory IO · PLC · pneumatic systems.

Robotics: FANUC RoboGuide · Raspberry Pi · Arduino · ESP32 · sensors and actuators.

Electronic Design (EDA): Proteus · KiCad · Multisim · LabVIEW · Micro-Cap.

Programming: MATLAB · Python · Arduino · C++.

Professional Skills

Technical problem-solving · Self-starter mindset · Project management · Working under pressure · Technical communication · Attention to detail · Teamwork.

Professional Experience

Brain Break Room, Automation and Systems Development – Puebla, Mexico Feb 2026 – Apr 2026

- Designed and built an ESP32-based interactive doll integrating a servo motor, voice output, and a laser proximity sensor to deliver automated clues inside an escape room.
- Developed and installed the electronic system of a wall-mounted piano with a custom PCB and, on my own initiative, added a sound amplification stage that noticeably improved audio quality.
- Handled the wiring, button assembly, and electrical integration of the interactive room systems.

Servicio de Administración Tributaria (SAT) — Mexican Tax Authority, Social Service – Cholula, Puebla Feb 2025 – Aug 2025

- Supported up to 5 taxpayers simultaneously during peak demand, sustaining service quality under pressure.
- Operated and troubleshooted the institution's information systems to validate and correct records in real time, reducing incidents and wait times.
- Managed and safeguarded sensitive taxpayer information and official documentation with full confidentiality.

MEVA & HAB Research Groups, Electronics Subcoordinator – Puebla, Mexico Aug 2022 – Dec 2024

- Programmed, calibrated, and integrated multiple sensors into an atmospheric probe for active-volcano monitoring, validating each with field tests prior to deployment.